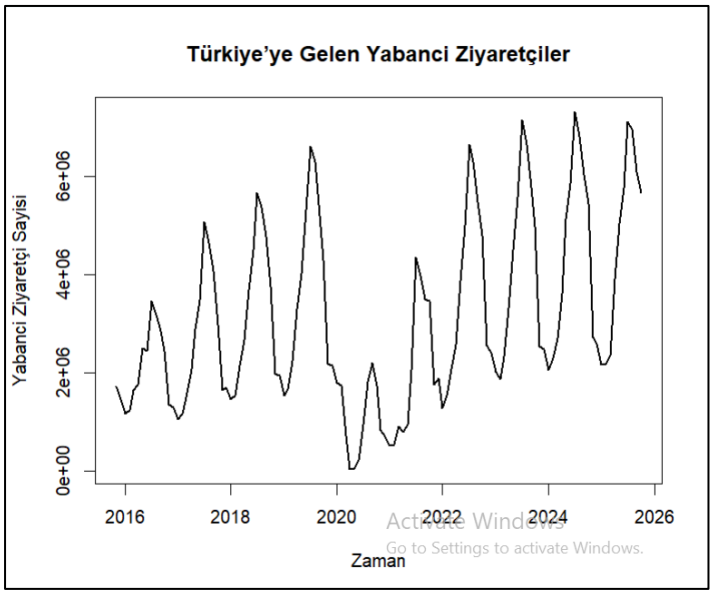
The dataset used in this study presents the number of foreign visitors arriving in Türkiye on a monthly basis. It covers the period from November 2015 to October 2025 and has a monthly frequency. Each observation in the series represents the total number of foreign visitors who arrived in Türkiye in the corresponding month. When the time series is examined, a long-term upward trend and a clear seasonal pattern that repeats regularly within each year can be observed.

Time Series Graph

```
rm(list = ls())

library(forecast)
library(fpp)
library(lmtest)
library(readxl)

View(data)
str(data)
data <- data[order(data$Year,
                   match(data$Month,
                         month.name)), ]
visitors_ts <- ts(data$`The number of foreign visitors to Turkey`,
                  start = c(2015,11),
                  frequency = 12)
ts.plot(visitors_ts,
        xlab = "Zaman",
        ylab = "Yabancı Ziyaretçi Sayısı",
        lwd = 2,
        main = "Türkiye'ye Gelen Yabancı Ziyaretçiler")
```



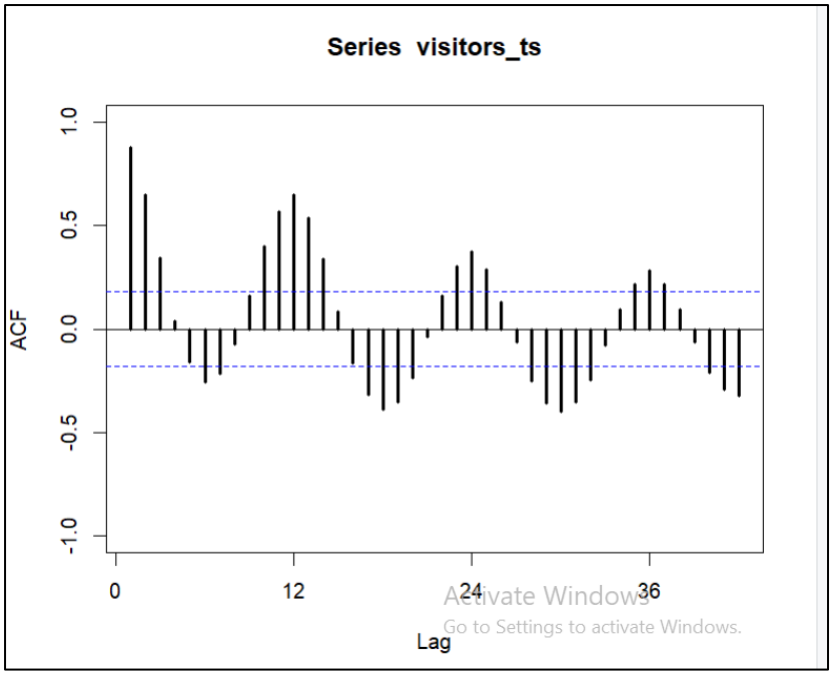
Graph1: Forign Vesitors coming to Turkey

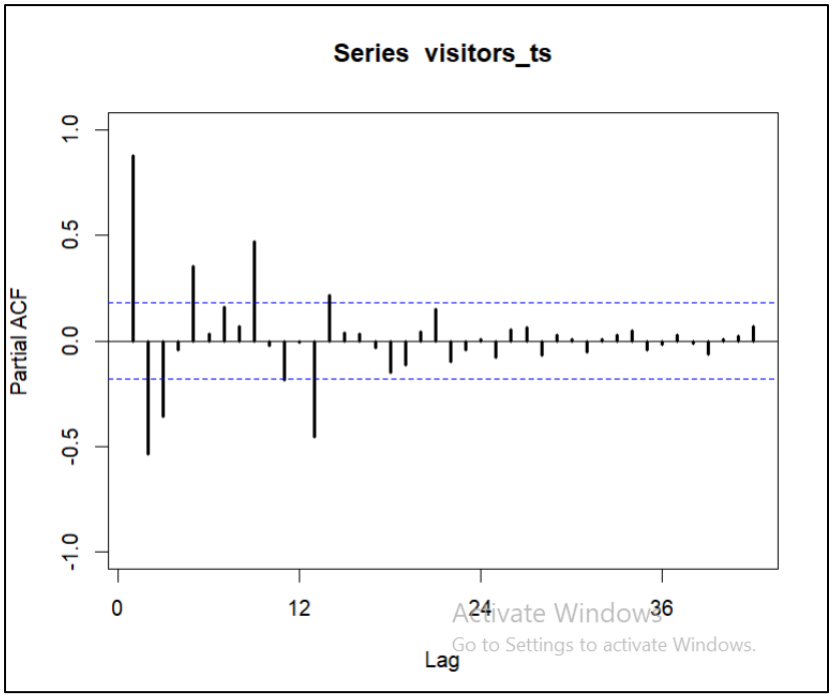
Interpretation

The time series plot shows the monthly variation in the number of foreign visitors arriving in Türkiye over time. When the graph is examined, it is observed that the series generally exhibits an upward trend. This indicates that, in the long run, the number of foreign visitors to Türkiye is increasing. In addition, there is a clear seasonal pattern in the series that repeats in specific months each year. It is observed that the number of visitors peaks during the summer months, while it decreases significantly during the winter months.

ACF &PACF Graphs:

```
Acf(visitors_ts, lag.max = 42, ylim = c(-1,1), lwd = 3)
Pacf(visitors_ts, lag.max = 42, ylim = c(-1,1), lwd = 3)
```





Interpretation

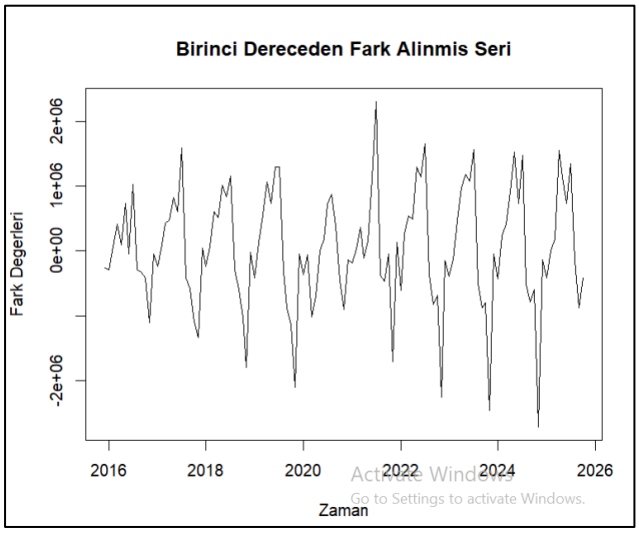
When the ACF (Autocorrelation Function) plot is examined, high and statistically significant positive autocorrelations are observed at the initial lags. The fact that the autocorrelation values decay slowly indicates that the series has a non-stationary structure and contains a clear trend component. In addition, the significant spikes at lags 12, 24, and 36 clearly show the presence of strong seasonality with a period of 12 (annual seasonality). This is an expected result for monthly tourism data.

In the PACF (Partial Autocorrelation Function) plot, a prominent and significant spike is observed especially at lag 1. After the first few lags, the partial autocorrelation values remain within the confidence bounds, suggesting that the series may have a low-order autoregressive (AR) structure. The limited spikes observed at seasonal lags (particularly around lag 12) indicate that a seasonal AR component may also be present in the series.

Based on the joint evaluation of the ACF and PACF plots, it can be concluded that the series contains both trend and seasonality, and therefore, it is non-stationary.

First-Order Differencing to Remove the Trend (d = 1)

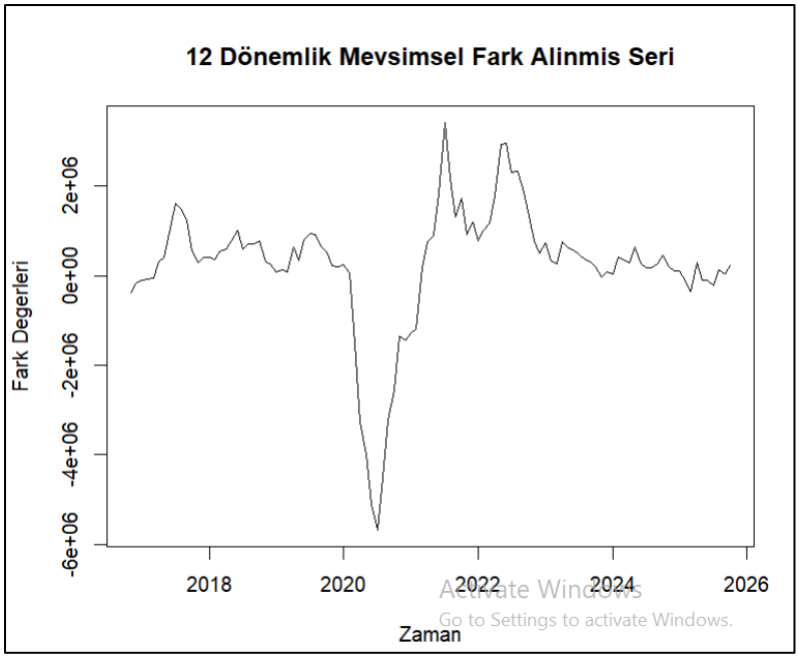
```
diff1 <- diff(visitors_ts, differences = 1)

ts.plot(diff1,
        main = "Birinci Dereceden Fark Alınmış Seri",
        ylab = "Fark Değerleri",
        xlab = "Zaman")
```



Grpah2: series with first degree difference

Seasonal Differencing

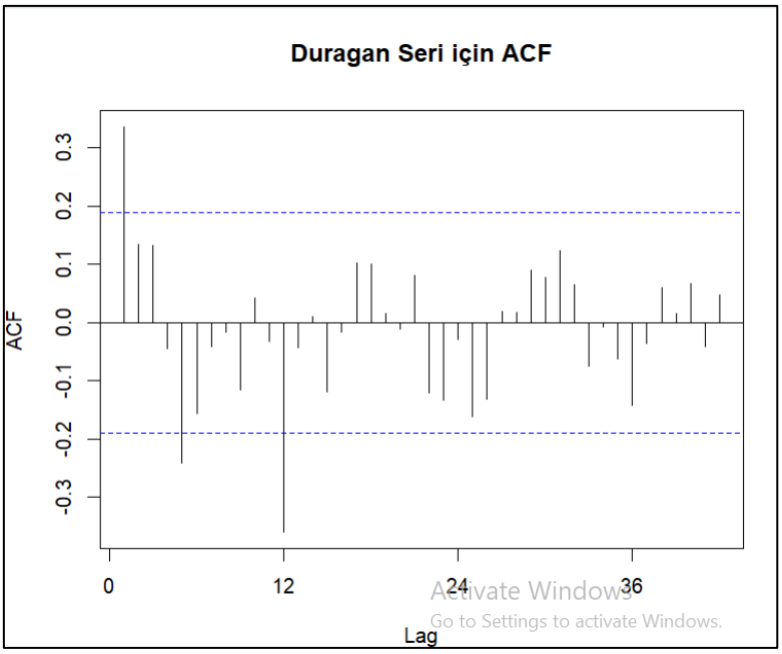
```
diff_seasonal <- diff(visitors_ts, lag = 12)

ts.plot(diff_seasonal,
        main = "12 Dönemlik Mevsimsel Fark Alınmış Seri",
        ylab = "Fark Değerleri",
        xlab = "Zaman")
```

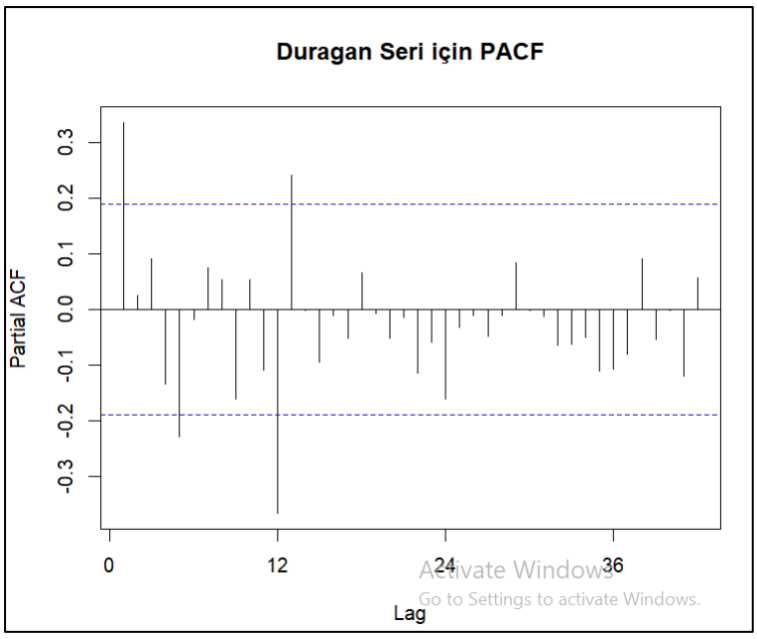


Graph3:The dataset obtained after applying a 12-period seasonal differencing

Stationarity Check (ACF & PACF)

```
Acf(diff_final,
    lag.max = 42,
    main = "Durağan Seri için ACF")

Pacf(diff_final,
     lag.max = 42,
     main = "Durağan Seri için PACF")
```



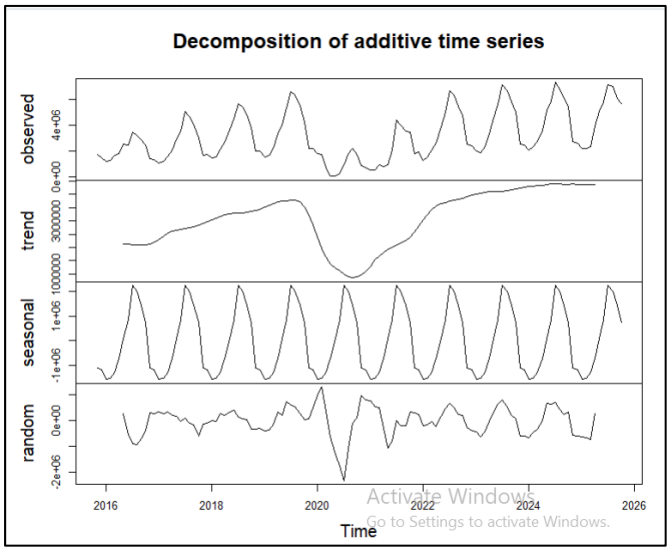


Interpretation

After applying first-order differencing and 12-period seasonal differencing, the ACF and PACF plots of the resulting series show that autocorrelations decay rapidly and significant correlations are limited to only the first few lags. This indicates that the series has been successfully cleared of trend and seasonal effects and has become stationary. Therefore, no further differencing is required.

Decomposition Methods

1- Additive Decomposition Method



Objective of the Model Within the scope of the additive decomposition method, the following linear regression model has been established to test whether there is a long-term trend component in the time series:

Additive Decomposition Model

H_0: β_1=0 (There is no significant trend in the series.

H_0: $\beta_1 \neq 0$ (There is significant trend in the series.

```
> summary(trend_add)

Call:
tslm(formula = visitors_ts ~ trend)

Residuals:
     Min       1Q   Median       3Q      Max 
-2988759 -1335632  -403768  1266562  3818804 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  1823062     318539   5.723 8.08e-08 ***
trend          21736       4569   4.757 5.61e-06 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1734000 on 118 degrees of freedom
Multiple R-squared:  0.1609,	Adjusted R-squared:  0.1538 
F-statistic: 22.63 on 1 and 118 DF,  p-value: 5.609e-06
```

Interpretation

According to the regression output obtained:

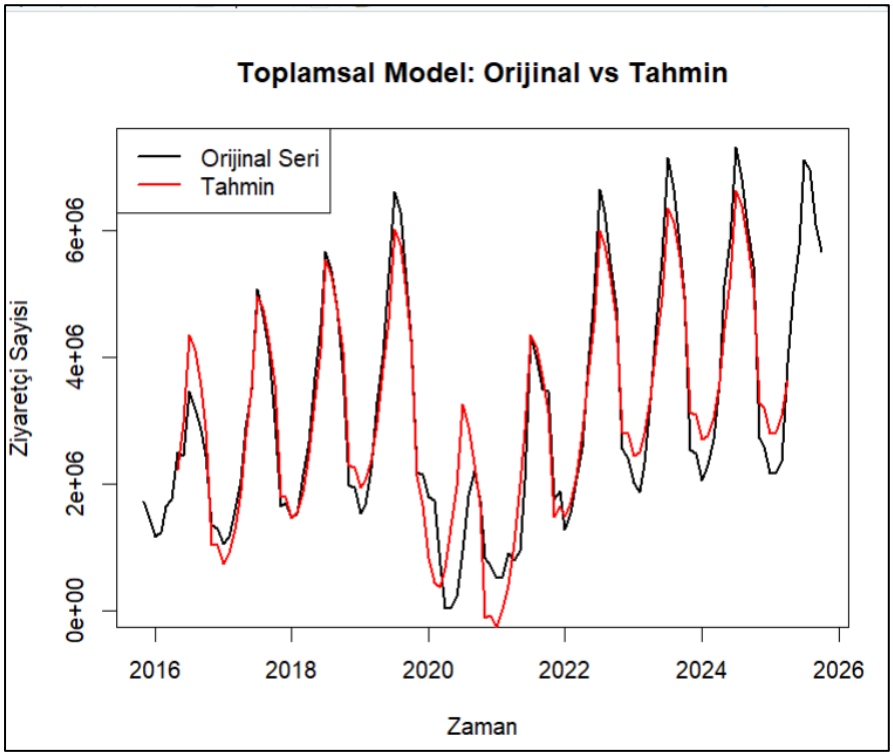
The trend coefficient (β_1) is estimated as 21.736.

The p-value associated with the trend coefficient is less than 0.001 (p = 5.61e-06).

This result indicates that the null hypothesis (H_0) is rejected at the 5% significance level.

Therefore, it is concluded that the number of foreign visitors to Türkiye exhibits a statistically significant and positive trend over time. In other words, as the years progress, an average increase in the number of foreign visitors is observed.

Assumption: Time Series Fit Between the Original Series and the Fitted Series

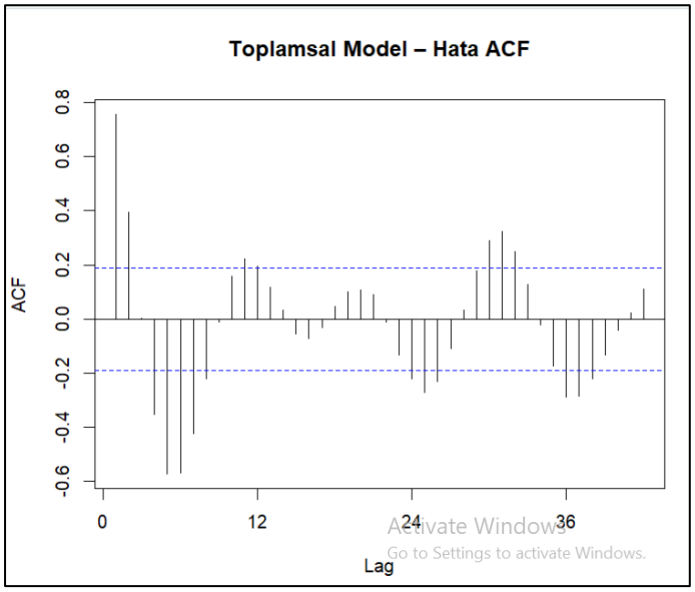


Interpretation

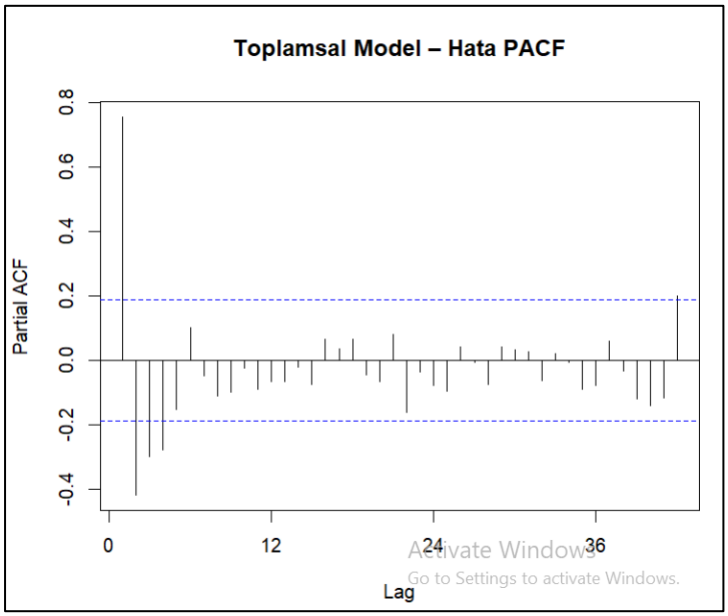
As seen from the graph, although the additive model is successful in capturing the overall trend and seasonal structure, the forecasts tend to remain below the original series, especially during periods when the series reaches high values. Since the amplitude of seasonal fluctuations is assumed to be constant over time, the model fails to fully reflect the increasing magnitude of fluctuations observed in later years.

While the fit between the predicted and actual series is better at lower levels, it is observed that forecast errors relatively increase as the level of the series rises. This suggests that the assumption of constant seasonal effects in the additive model is somewhat restrictive for this dataset.

Assumption: The Error Series is White Noise



ACF & PACF graphs for the error series




```
> Box.test(add_decomp$random,
+          lag = 24,
+          type = "Ljung-Box")

        Box-Ljung test

data:  add_decomp$random
X-squared = 228.15, df = 24, p-value < 2.2e-16

>
```

Interpretation

When examining the ACF plot of the residuals from the additive model: Autocorrelation coefficients at many lags (especially the first few lags and around seasonal lags) exceed the confidence bounds. This indicates that the residuals are not random (i.e., not white noise). In particular, the presence of clear patterns around lag 12 suggests that seasonal effects have not been fully captured by the model.

Interpretation: The additive decomposition model has failed to fully explain the dependency structure in the series.

PACF Plot of Residuals: A very high and significant partial autocorrelation is observed at the first lag. There are also values exceeding the confidence bounds at other lags. This structure suggests that there is a remaining AR-type dependency among the residuals.

Interpretation: The presence of structure in the residuals after modeling indicates that the model is inadequate.

Ljung–Box Test The results of the Ljung–Box test are as follows:

Test statistic: $\chi^2 = 228.15$

Degrees of freedom: 24

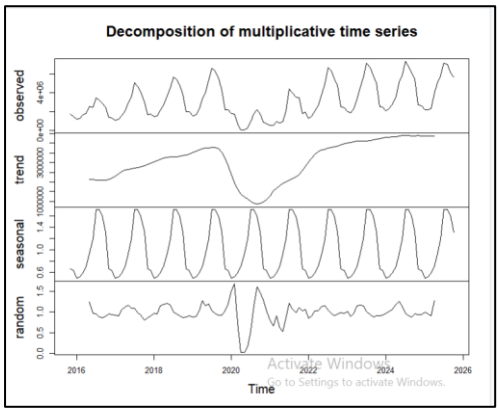
p-value $< 2.2 \times 10^{-16}$

Based on these results: The null hypothesis (H_0: residuals are independent) is rejected. There is significant autocorrelation among the residuals.

Interpretation: The residuals of the additive decomposition model do not exhibit white noise properties.

Overall Conclusion: When all findings are considered together, the additive decomposition model has not been sufficient to fully capture the trend and seasonal structure of the series.

<u>Multiplicative Decomposition Model</u>



Objective of the Model

Within the scope of the multiplicative decomposition method, the following linear regression model has been established to test whether there is a long-term trend component in the foreign visitors series adjusted for seasonal effects:

Multiplicative Decomposition Model

$H_0 : \beta_1 = 0$ (There is no significant trend in the seasonally adjusted series.)

$H_1 : \beta_1 \neq 0$ (There is significant trend in the seasonally adjusted series.)

```
> summary(trend_mult)

Call:
tslm(formula = visitors_ts/mult_decomp$seasonal ~ trend)

Residuals:
     Min       1Q   Median       3Q      Max 
-3007287  -197997   225228   543466  1906093 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2043580     187660  10.890  < 2e-16 ***
trend          18103       2692   6.725 6.62e-10 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1021000 on 118 degrees of freedom
Multiple R-squared:  0.2771,	Adjusted R-squared:  0.271 
F-statistic: 45.23 on 1 and 118 DF,  p-value: 6.622e-10

>
```

Interpretation According to the regression output:

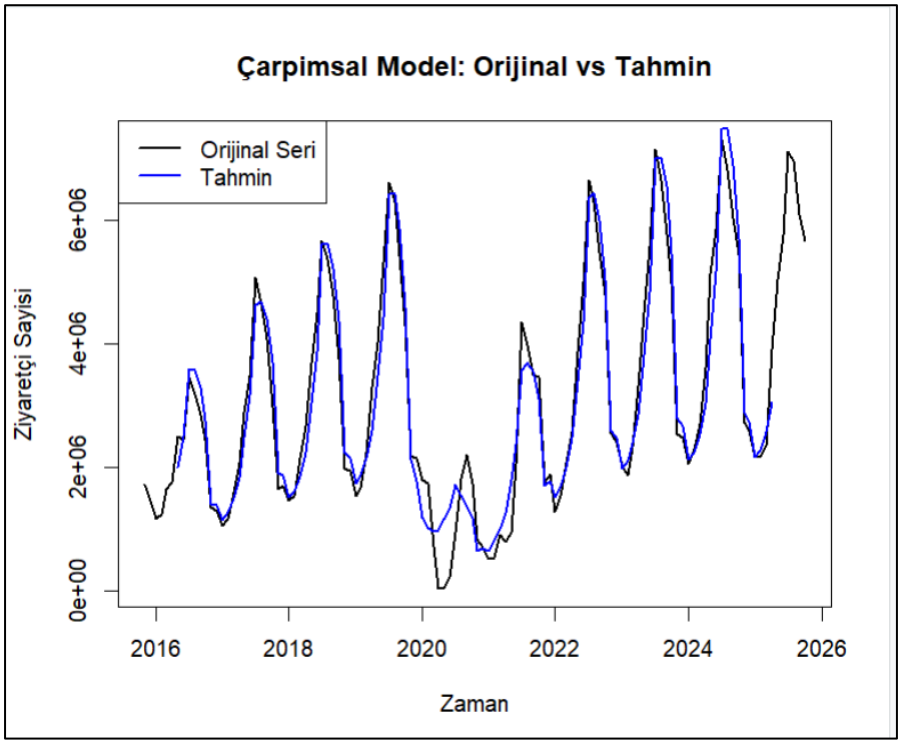
The trend coefficient (β_1) is estimated as 18.103.

The p-value associated with the trend coefficient is less than 0.001 (p = 6.62e-10).

Based on this result, the null hypothesis (H_0) is rejected at the 5% significance level.

Therefore, it is concluded that there is a statistically significant and positive trend in the seasonally adjusted foreign visitor series. As time progresses, the number of visitors shows a long-term increasing tendency.

Assumption: Time Series Fit Between the Original Series and the Fitted Series

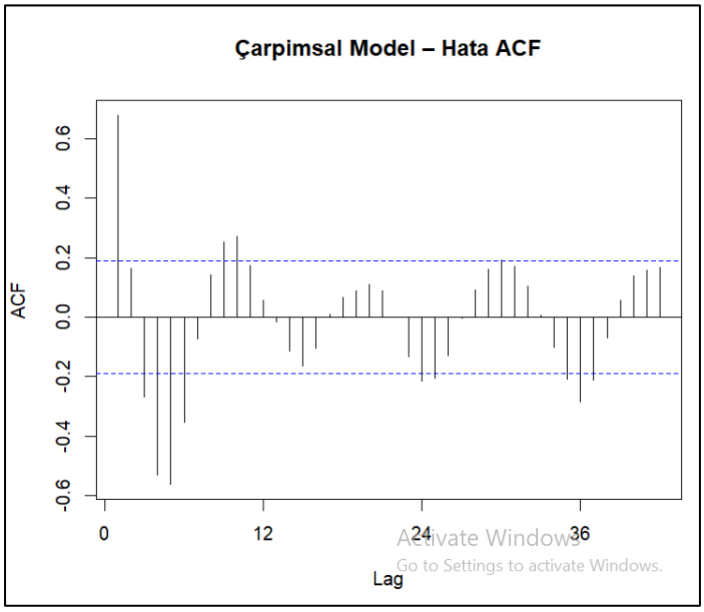


Interpretation

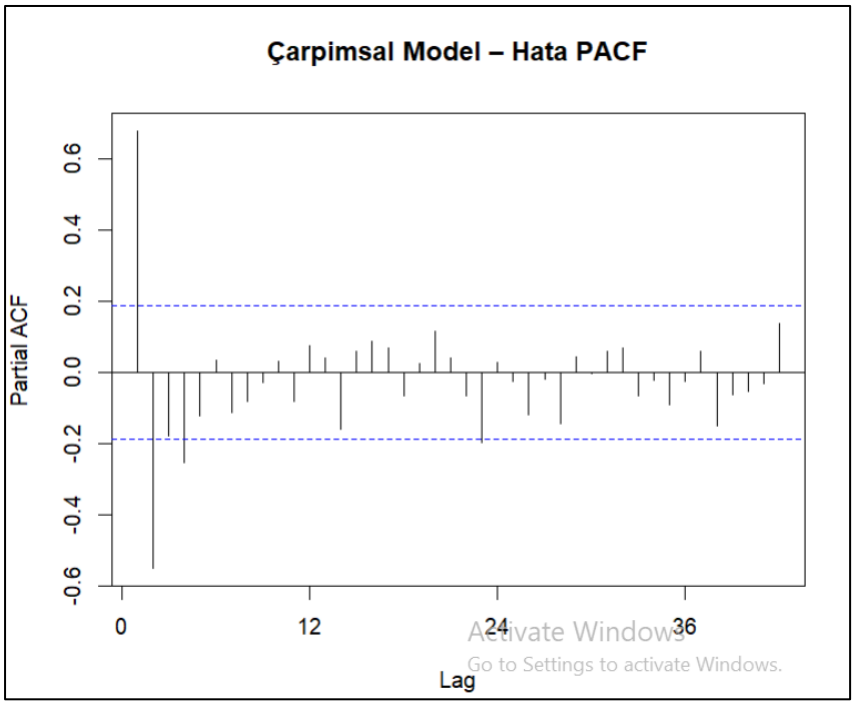
When the graph is examined, it is observed that the fitted series (blue line) closely follows the original series (black line) in terms of both trend and seasonal fluctuations. Especially at the peak and trough points of the series, the amplitude of the fitted values is largely consistent with the variations in the original series. This indicates that the seasonal effect changes depending on the increasing level over time and that the multiplicative model successfully captures this structure.

Although there are minor deviations during structural breaks, such as the sharp decline observed around 2020, the model is generally successful in explaining the dynamics of the series. The limited differences between the fitted and original series suggest that the multiplicative decomposition model is appropriate for the data structure.

Assumption: The Error Series is White Noise



ACF&PACF for the error series




```
> Box.test(mult_decomp$random,
+          lag = 24,
+          type = "Ljung-Box")

        Box-Ljung test

data:  mult_decomp$random
X-squared = 189.17, df = 24, p-value < 2.2e-16
```

Interpretation

When examining the ACF plot of the residuals from the multiplicative model: Negative autocorrelations exceeding the confidence bounds are observed at the initial lags (especially around lags 1–3). Some significant spikes are still present around seasonal lags (≈ 12 and its multiples). However, these spikes are weaker and more limited compared to the additive model.

Interpretation: Although the multiplicative model explains part of the seasonal structure, the residuals are not completely random.

PACF Plot of Residuals: There is a clear and significant negative partial autocorrelation at the first lag. At subsequent lags, most values remain within the confidence bounds. Compared to the additive model, the PACF structure decays more rapidly.

Interpretation: This indicates that the multiplicative model captures the dependency structure better than the additive model, but does not eliminate it entirely.

Ljung–Box Test:

$\chi^2 = 189.17$

df = 24

p-value $< 2.2 \times 10^{-16}$

Based on these results: The null hypothesis (H_0: residuals are independent) is rejected. There is still significant autocorrelation among the residuals.

Interpretation: The residuals of the multiplicative decomposition model do not exhibit white noise properties.

Overall Conclusion: Compared to the additive model, weaker autocorrelations are observed in the ACF and PACF plots of the residuals obtained after the multiplicative decomposition. Additionally, the lower Ljung–Box test statistic suggests that the multiplicative model explains the seasonal structure of the series better. However, since the residuals do not fully exhibit white noise properties, the model is still not sufficient to completely explain the series.

General Review for Decomposition Methods

Conclusion

In this study, the time series of the number of visitors was analyzed using additive and multiplicative decomposition methods in order to reveal the structural components of the series. During the decomposition process, the series was separated into trend, seasonal, and random (error) components.

The findings indicate the presence of a clear long-term upward trend in the series. The trend component was estimated using linear regression in both models, and the trend coefficient was found to be statistically significant. This suggests a systematic increase in the number of visitors over time.

When the seasonal component was examined, it was determined that the series has a regular and strong seasonal structure. It was observed that the magnitude of seasonal fluctuations changes depending on the level of the series; this indicates that the multiplicative decomposition model better represents the data structure. Indeed, the fitted series obtained from the multiplicative model follows the original series more closely in terms of both level and fluctuation magnitude.

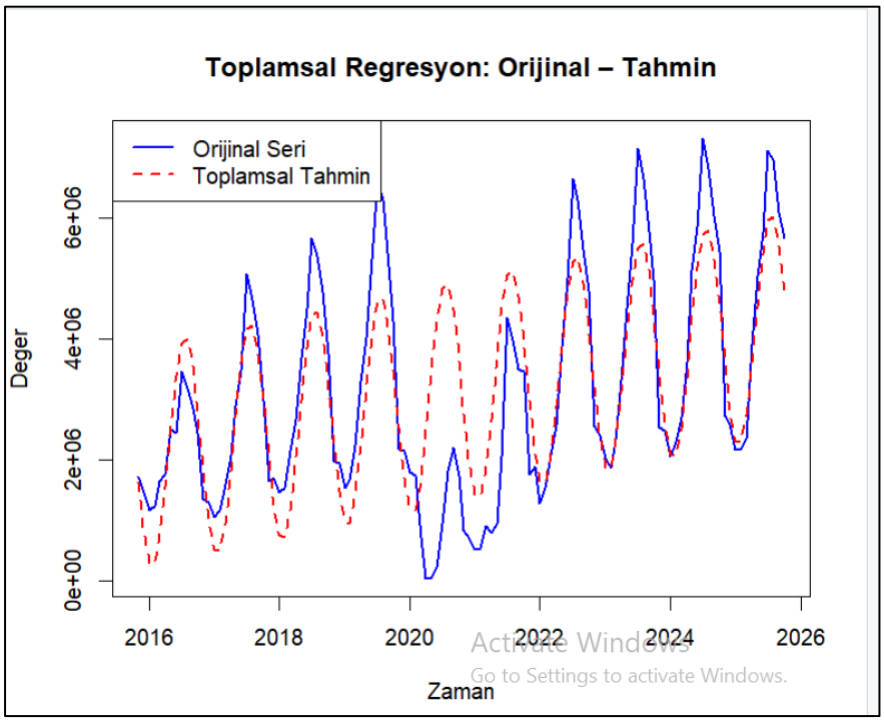
The ACF, PACF, and Ljung–Box test results of the residual (error) series show that, in both models, the residuals do not fully exhibit white noise properties. However, considering that the primary objective of decomposition methods is not to explain all dependencies in the series but rather to separate its structural components, these results do not undermine the validity of the methods.

Overall, while both decomposition methods reveal the fundamental structure of the series, it is concluded that the multiplicative decomposition model has a stronger ability to explain the trend and seasonal components. Therefore, in the continuation of the study, it is considered appropriate to proceed with regression analysis based on the insights obtained from the decomposition results.

Regression Analysis

Aditive Regression

Original vs Fitted Series Fit for the Additive Model



Interpretation

It is observed that the fitted series largely follows the overall trend and seasonal structure of the original time series. The timing of peaks and troughs mostly coincides, indicating that the model successfully captures the main deterministic components of the series.

However, especially during periods of sudden fluctuations, the predicted values deviate noticeably from the observed values. These deviations suggest that short-term dynamics and the random component are not adequately represented by the model.

Conclusion:

The model provides a good structural fit in terms of trend and seasonality; however, these shortcomings lead to the presence of serial correlation in the residuals.

The statistical significance of the trend coefficient was tested under the following hypotheses:

```
> summary(model_add)

Call:
lm(formula = y ~ t + sin1 + cos1, data = data_add)

Residuals:
     Min       1Q   Median       3Q      Max 
-4080460  -406530   266979   709113  2009382 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  2003339     209362   9.569 2.43e-16 ***
t              18757       3005   6.241 7.34e-09 ***
sin1        -1763236     147200 -11.979  < 2e-16 ***
cos1          570502     146802   3.886  0.00017 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1137000 on 116 degrees of freedom
Multiple R-squared:  0.6453,	Adjusted R-squared:  0.6362 
F-statistic: 70.36 on 3 and 116 DF,  p-value: < 2.2e-16
```

Interpretation

Overall Structure of the Model The additive regression model is defined as: y

$y=\beta_0 + \beta_1 t + \beta_2 \sin(\cdot) + \beta_3 \cos(\cdot) + \ldots$ and aims to jointly explain the linear trend and seasonal structure in the time series.

Numerical Interpretation of Coefficients

The intercept is 2,003,339 and is highly statistically significant ($t = 9.569$, $p = 2.43e{-}16$). This indicates that the average initial level of the series is relatively high.

The coefficient of the time variable is 18.757 and is strongly significant ($t = 6.241$, $p = 7.34e{-}09$). Accordingly, when time increases by one unit, the number of visitors increases on average by approximately 18,757. This confirms the presence of a clear upward trend in the series.

The coefficient of the sine term is −1,763,236 and is highly significant ($t = -11.979$, $p < 2e{-}16$). This indicates that during certain periods of the seasonal cycle, the number of visitors falls approximately 1.76 million below the average.

The coefficient of the cosine term is 570,502 and is statistically significant ($t = 3.886$, $p = 0.00017$). The joint significance of the sine and cosine terms clearly reveals a strong and regular annual seasonality in the series.

Model Fit and Explanatory Power

The coefficient of determination is R 2

R 2 =0.6453, meaning that approximately 64.5% of the total variation in visitor numbers is explained by the model.

The adjusted R 2 R 2 is 0.6362, indicating that the model maintains a high explanatory power despite the number of predictors.

The F-statistic is 70.36 with a p-value < 2.2e−16, showing that the model is overall highly statistically significant.

Residuals and Overall Evaluation

The standard error of the residuals is 1,137,000. While this is expected given the large scale and variability of the series, it also indicates that the model does not fully capture short-term fluctuations.

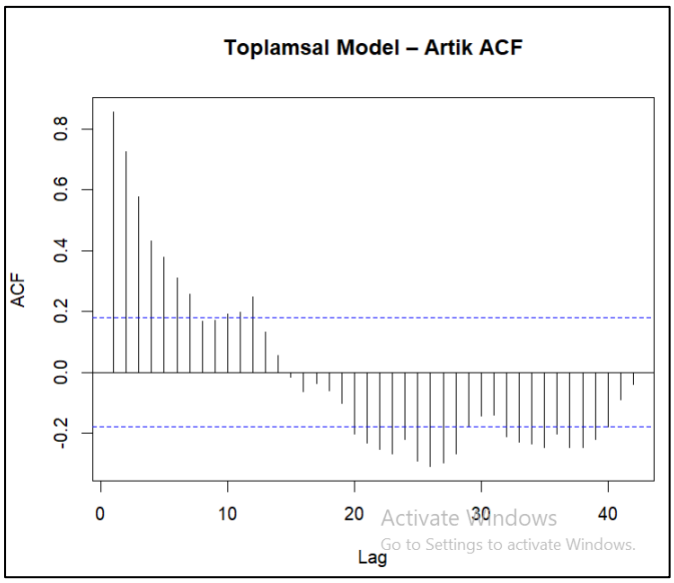
Overall Conclusion The additive regression model successfully captures:

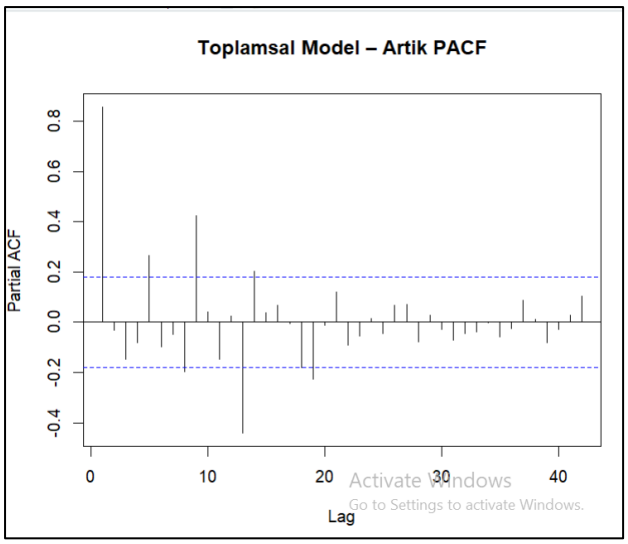
a strong positive trend ($\beta_1 = 18.757$),

a clear seasonal structure ($\beta_2 = -1{,}763{,}236$, $\beta_3 = 570{,}502$).

The model is effective in explaining the long-term behavior of the series; however, the relatively high residual variance suggests that prediction accuracy may be limited, especially at peak and trough points.

Whether the Error Series is White Noise:






```
> Box.test(res_add, lag = 24, type = "Ljung")

        Box-Ljung test

data:  res_add
X-squared = 333.31, df = 24, p-value < 2.2e-16
```

```
> library(lmtest)
> dwtest(model_add)

        Durbin-Watson test

data:  model_add
DW = 0.28134, p-value < 2.2e-16
alternative hypothesis: true autocorrelation is greater than 0
```

Interpretation

Residual ACF Plot In the ACF plot of the residuals, autocorrelation coefficients exceed the 95% confidence bounds at many lags. Strong positive autocorrelation is especially evident at the first few lags. At higher lags, this effect gradually decreases and transitions into negative autocorrelation. This pattern indicates that the residuals are not independent over time.

Conclusion: The presence of significant autocorrelation among residuals shows that the additive regression model fails to fully capture the dependency structure of the time series. Therefore, the residuals do not satisfy the white noise assumption.

Residual PACF Plot In the PACF plot, statistically significant spikes are observed particularly at the first lag and at some higher lags. This suggests that even after fitting the additive regression model, an autoregressive structure remains in the residuals.

Conclusion: The presence of significant lags in the PACF indicates that the residuals are not white noise. This implies that additional dynamic components (such as AR terms or ARIMA-based error structures) should be incorporated into the model.

Ljung–Box Test

$\chi^2 = 333.31$

df = 24

p-value $< 2.2 \times 10^{-16}$

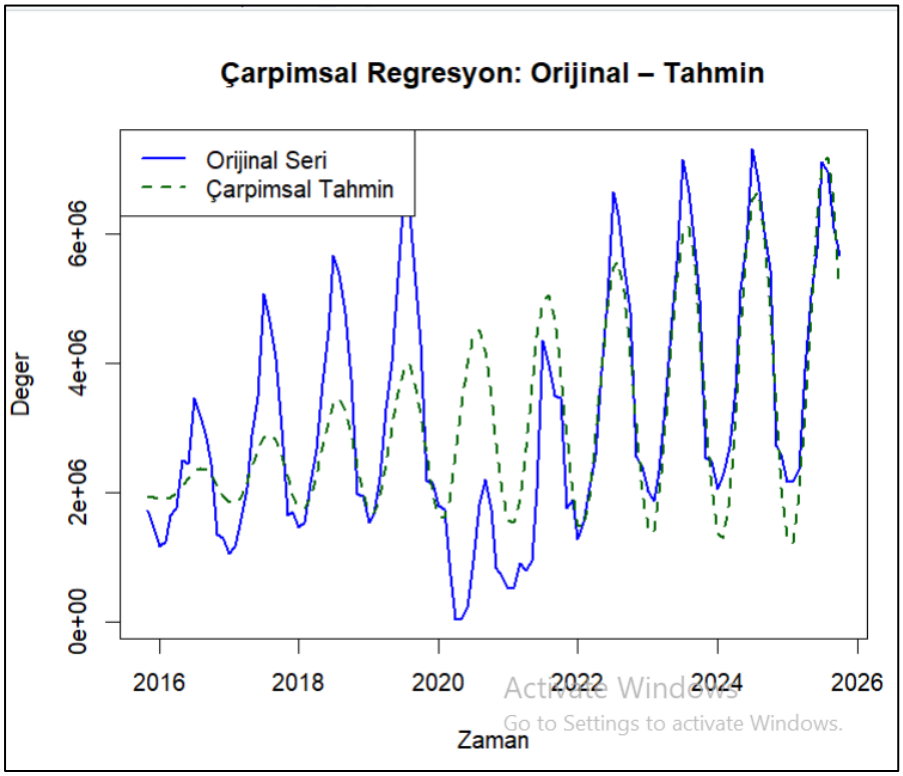
This result leads to the rejection of the null hypothesis that the residuals are independent up to a certain lag.

Conclusion: The Ljung–Box test statistically confirms the presence of strong autocorrelation among the residuals, clearly indicating that they are not white noise.

Durbin–Watson Test The Durbin–Watson statistic is extremely low (DW = 0.281), and the p-value is highly significant ($p < 2.2e{-}16$). This indicates the presence of strong first-order positive autocorrelation in the regression residuals.

<u>Multiplicative regression</u>

Original vs Fitted Series Fit for the Multiplicative Model



Interpretation

Fit Between the Fitted Series and the Original Series
When the original series and the multiplicative regression forecasts are examined together, it is observed that the model successfully captures the overall trend and seasonal structure. In particular, the increasing trend and periodic fluctuations are clearly represented in the fitted series.

However, there are noticeable deviations at peak and trough points. In some periods, the model underestimates the peaks, while in others it overestimates the troughs. This indicates that although the model is successful in explaining the general structure, it is not able to fully capture short-term fluctuations and sudden changes.

The statistical significance of the trend coefficient was tested under the following hypotheses:

```
> summary(model_mult)

Call:
lm(formula = y ~ t + s1 + c1, data = data_mult)

Residuals:
     Min       1Q   Median       3Q      Max 
-3773658  -483879   153288   615778  2710690 

Coefficients:
            Estimate Std. Error t value Pr(>|t|)    
(Intercept)  1926618     214244   8.993 5.41e-15 ***
t              19193       3078   6.236 7.53e-09 ***
s1            -24867       2178 -11.416  < 2e-16 ***
c1              7812       2145   3.642 0.000406 ***
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

Residual standard error: 1165000 on 116 degrees of freedom
Multiple R-squared:  0.6275,	Adjusted R-squared:  0.6178 
F-statistic: 65.13 on 3 and 116 DF,  p-value: < 2.2e-16
```

Interpretation

Interpretation of Coefficients

Intercept (β_0 = 1,926,618): Represents the baseline level of the series when time and seasonal effects are zero.

Time coefficient (β_1 = 19,193): Indicates a statistically significant increase in the series over time.

s_1 (β_2 = −24,867): Represents the negative contribution of the seasonal component.

c_1 (β_3 = 7,812): Represents the complementary positive effect of the seasonal cycle.

Statistical Significance For all coefficients, the p-value is < 0.001. This indicates that both the trend and seasonal components are statistically significant.

Model Explanatory Power

R 2 =0.6275

Adjusted R 2

R 2 =0.6178

Interpretation: The model explains approximately 63% of the total variance in the dependent variable. The closeness of adjusted R 2 R 2 to R 2 R 2 suggests that the included variables contribute meaningfully to the model and that overfitting is not a concern.

Overall Model Significance

F-statistic = 65.13

p-value < 2.2e−16

Interpretation: According to the F-test, the model is statistically significant overall. This means that the independent variables, when considered together, have a significant effect on the dependent variable.

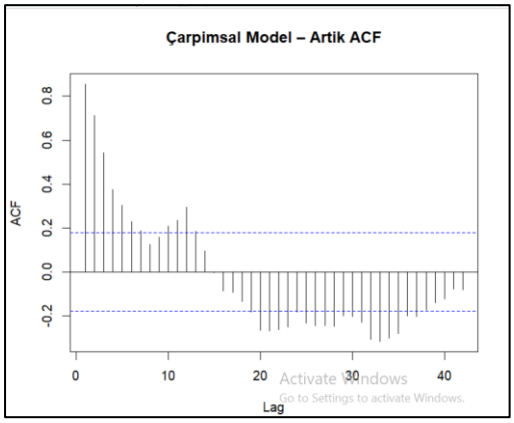
Residual Standard Error

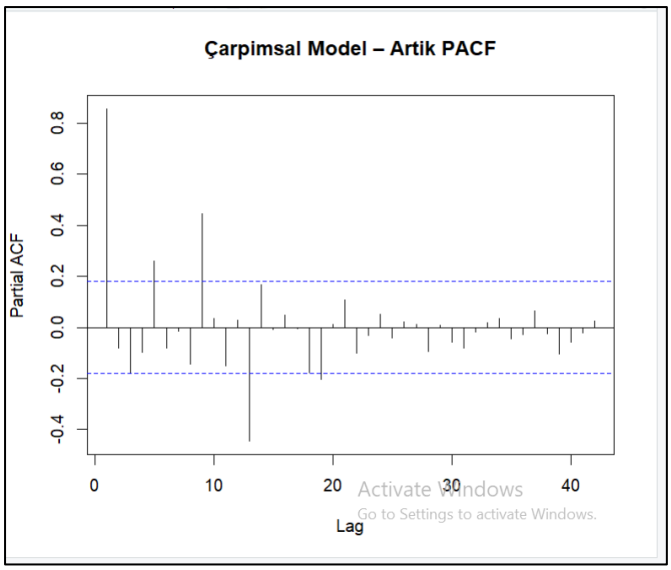
Residual standard error = 1,165,000

Interpretation: This value indicates that, on average, the model's predictions deviate from the actual values by approximately 1.16 million units. This reflects the large scale and variability of the dataset.

Overall Conclusion The multiplicative regression model successfully captures the trend and seasonal components of the time series in a statistically significant manner and explains a substantial portion of the variance in the dependent variable. Although the model is generally significant, the magnitude of the residuals suggests that the series has high variability and that comparison with more advanced time series models may be necessary.

Whether the Error Series is White Noise






```
> Box.test(res_mult, lag = 24, type = "Ljung")

        Box-Ljung test

data:  res_mult
X-squared = 322.04, df = 24, p-value < 2.2e-16
```

```
> dwtest(model_mult)

        Durbin-Watson test

data:  model_mult
DW = 0.28576, p-value < 2.2e-16
alternative hypothesis: true autocorrelation is greater than 0
```

Interpretation

White Noise (Residuals) Assumption

ACF Plot In the ACF plot of the residuals, autocorrelations exceeding the confidence bounds are observed at multiple lags. Interpretation: There is significant dependence among the residuals, indicating that they are not randomly distributed.

PACF Plot In the PACF plot, clear and statistically significant spikes are observed, especially at the first few lags. Interpretation: A remaining structure exists in the residual series, meaning the model has not fully captured all dependency patterns in the data.

Ljung–Box Test The Ljung–Box test yields a p-value < 2.2e−16. Interpretation: The null hypothesis (H_0) is rejected. The residuals are not independent, and the white noise assumption is not satisfied.

Durbin–Watson Test The Durbin–Watson statistic is approximately 0.29, with a very small p-value. Interpretation: There is strong positive autocorrelation in the residuals. This result supports the findings from the ACF and PACF plots.

Exponential Smoothing Method
Simple Exponential Smoothing

```
> SES <- ets(visitors_ts, model = "ANN")
> summary(SES)
ETS(A,N,N)

Call:
ets(y = visitors_ts, model = "ANN")

  Smoothing parameters:
    alpha = 0.9999

  Initial states:
    l = 2056997.1537

  sigma:  898369.2

     AIC     AICc      BIC
3868.483 3868.690 3876.845

Training set error measures:
                    ME     RMSE      MAE       MPE     MAPE      MASE     ACF1
Training set 30195.05 890851.3 684337.1 -26.6477 49.63044 0.7371822 0.466486
```

Interpretation

Model Structure and Smoothing Parameter The Simple Exponential Smoothing (SES) model includes only a level component and does not account for trend or seasonality.

α (alpha) = 0.9999

Interpretation: The fact that the alpha parameter is very close to 1 indicates that the model assigns almost all the weight to the most recent observation. This suggests that the series exhibits high variability and structural fluctuations over time, and that the model is not effectively utilizing past information.

Model Fit Criteria (AIC, AICc, BIC)

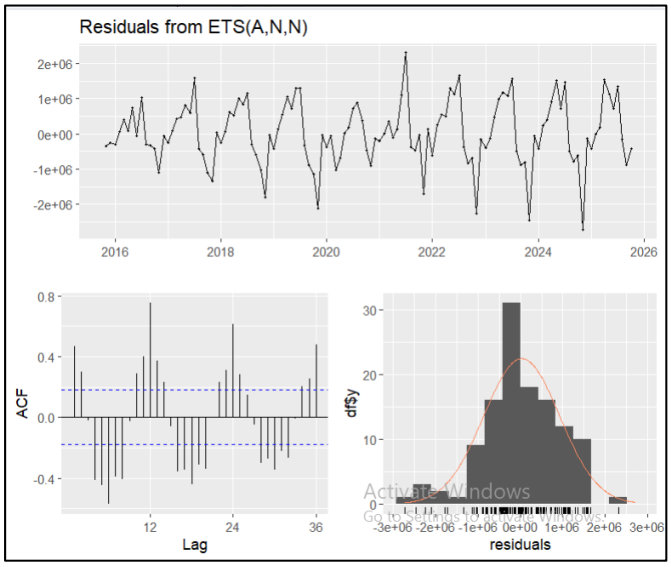
AIC = 3868.483

AICc = 3868.690

BIC = 3876.845

Interpretation: These values are relatively high compared to more advanced models such as Holt and Holt–Winters. This indicates that the Simple Exponential Smoothing model provides a poor fit for the data.

```
> checkresiduals(SES, lag = 42)

        Ljung-Box test

data:  Residuals from ETS(A,N,N)
Q* = 662.04, df = 42, p-value < 2.2e-16

Model df: 0.   Total lags used: 42
```



Interpretation

Ljung–Box Test Results

Q = 662.04*

Degrees of freedom (df) = 42

p-value < 2.2e−16

Interpretation: The null hypothesis (H_0: residuals are independent) is rejected. This indicates that the residuals are not white noise and that the model fails to capture the temporal dependence in the series.

ACF Plot of Residuals Observations:

In the ACF plot, the 95% confidence bounds are exceeded at many lags.

Strong autocorrelations are especially evident at seasonal lags.

Interpretation: The ACF plot shows that a strong and regular dependency structure remains in the series. This indicates that the Simple Exponential Smoothing model fails to capture both the trend and seasonality.

Overall Conclusion The Simple Exponential Smoothing (ETS(A,N,N)) model does not account for either trend or seasonal components, resulting in strong autocorrelations in the residuals. The Ljung–Box test and ACF results clearly show that the white noise assumption is violated, indicating that this model is not appropriate for the given dataset.

Holt's Exponential Smoothing

```
> summary(Holt)
ETS(A,Ad,N)

Call:
ets(y = visitors_ts, model = "AAN")

  Smoothing parameters:
    alpha = 0.8395
    beta  = 0.8395
    phi   = 0.8037

  Initial states:
    l = 891000.5857
    b = 226728.7153

  sigma:  819038.6

     AIC     AICc      BIC
3849.205 3849.948 3865.930

Training set error measures:
                    ME     RMSE      MAE      MPE     MAPE      MASE         ACF1
Training set 2637.745 801793.8 652211.1 15.11145 50.70905 0.7025754 0.000957894
```

Interpretation

Interpretation of Smoothing Parameters

α = 0.8395: The level component is updated rapidly. The model is highly responsive to new observations.

β = 0.8395: The trend component is also updated quickly, indicating that the trend changes significantly over time.

φ = 0.8037: The trend is damped, meaning its effect decreases in the long run.

Interpretation: The model assigns strong weights to both level and trend components, allowing it to respond quickly to short-term changes. However, since seasonality is not included, these responses may sometimes be misleading.

Initial States

Initial level (l) ≈ 891,001

Initial trend (b) ≈ 226,729

Interpretation: The series starts with a strong upward trend. However, part of this increase may actually be driven by seasonal fluctuations.

Model Error (Sigma)

σ = 819,038.6

Interpretation: The high standard deviation of errors indicates that the model fails to adequately capture the variability in the data. This is largely due to the exclusion of seasonality.

Information Criteria (Model Comparison)

AIC = 3849.205

AICc = 3849.948

BIC = 3865.930

Interpretation: These values are higher compared to Holt–Winters additive and multiplicative models, indicating that the Holt model provides a weaker fit for this dataset.

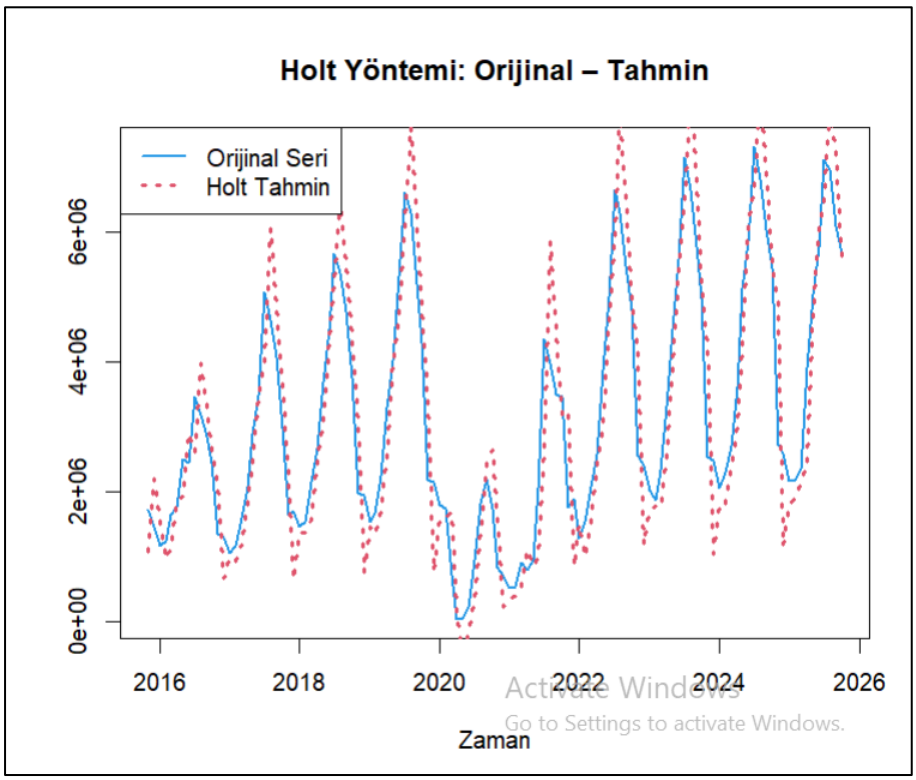
Forecast Performance (Key Metrics)

MAPE = 50.71%: Indicates a high relative error.

MASE = 0.703: Better than the naïve model, but weaker compared to Holt–Winters models.

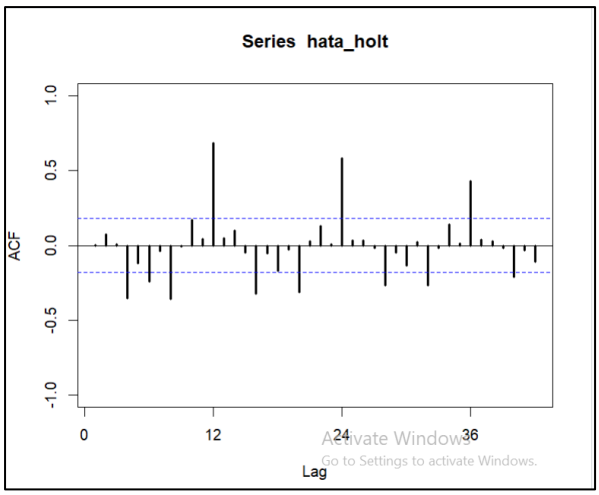
RMSE = 801,794: Indicates large forecast errors.

Interpretation: The forecasting performance is clearly inferior compared to models that incorporate seasonality.

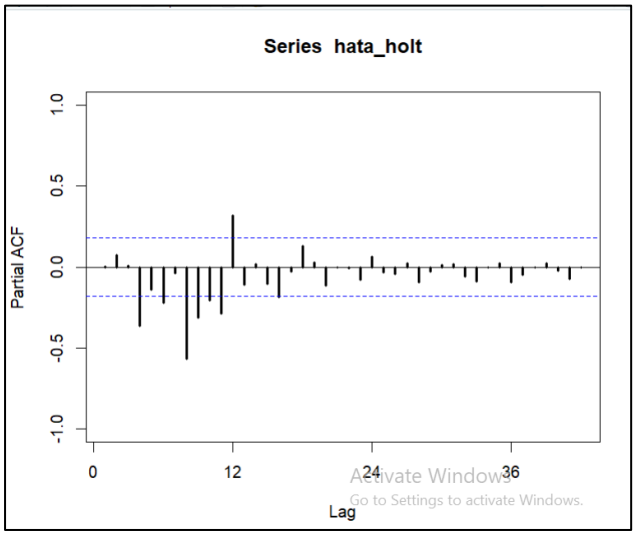


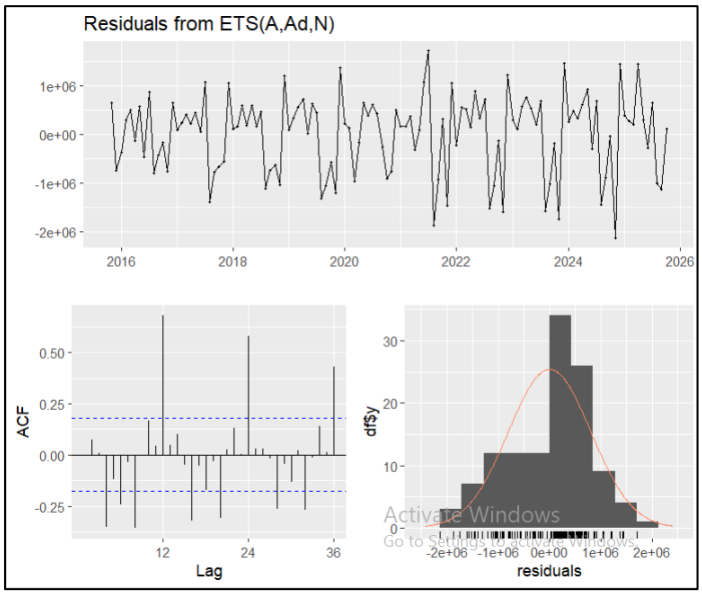
Fit between Holt estimation and original series

```
> Box.test(hata_holt, lag = 24, type = "Ljung")

        Box-Ljung test

data:  hata_holt
X-squared = 198.99, df = 24, p-value < 2.2e-16
```



ACF&PACF for the error series





Interpretation

Fit Between Original and Fitted Series When the graphs are examined, it is observed that Holt's forecasts capture the overall trend of the series; however, they fail to explain the strong seasonal fluctuations. This leads to noticeable forecast errors, especially at peak and trough points.

White Noise Assumption

ACF1 = 0.001: Autocorrelation at the first lag is very low.

Ljung–Box test (lag = 24): p-value < 2.2e−16

Interpretation: Although the ACF1 value is low, the significant Ljung–Box test indicates that the residuals are not independent overall. Some significant deviations are also observed in the ACF and PACF plots. Therefore, the white noise assumption is not satisfied.

Evaluation of ACF and PACF Based on the joint analysis of ACF and PACF plots:

There is statistically significant autocorrelation in the residuals.

The residuals do not exhibit white noise behavior.

These findings are consistent with the significant Ljung–Box test result (p < 2.2e−16).

Overall Interpretation According to the ACF and PACF plots, dependency structure remains in the residuals of the Holt exponential smoothing model. This indicates that the model is insufficient in capturing the seasonal structure and all dynamics of the series.

Holt–Winters Additive Method

```
> summary(winters_add)
ETS(A,Ad,A)

Call:
ets(y = visitors_ts, model = "AAA")

  Smoothing parameters:
    alpha = 0.9948
    beta  = 0.1412
    gamma = 0.0043
    phi   = 0.8006

  Initial states:
    l = 2463689.654
    b = 10201.5804
    s = 743877.4 1451541 2005750 2255020 842880.2 90532.91
           -734780.5 -1252267 -1532062 -1567391 -1169619 -1133483

  sigma:  421053.8

     AIC     AICc      BIC
3700.291 3707.064 3750.466

Training set error measures:
                   ME     RMSE      MAE       MPE     MAPE      MASE      ACF1
Training set 13645.16 390090.7 274484.7 -53.25629 66.04825 0.2956806 0.2522586
>
```

Interpretation

Key Smoothing Parameters

$\alpha = 0.9948$: The model assigns very high weight to recent observations, indicating that the series is volatile and sensitive to sudden changes.

$\gamma = 0.0043$: The seasonal component remains almost constant over time; seasonal patterns are stable across years.

$\varphi = 0.8006$: The trend is damped, meaning its effect decreases in the long run. This helps produce more realistic long-term forecasts.

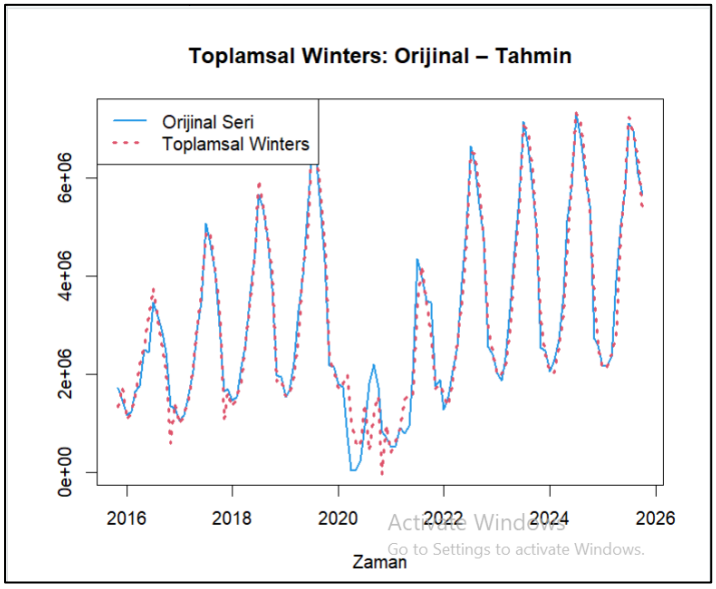
Initial Structure The series has a high initial level and a positive trend tendency. The presence of both positive and negative seasonal components indicates strong and clear seasonality.

Model Performance and Error Structure

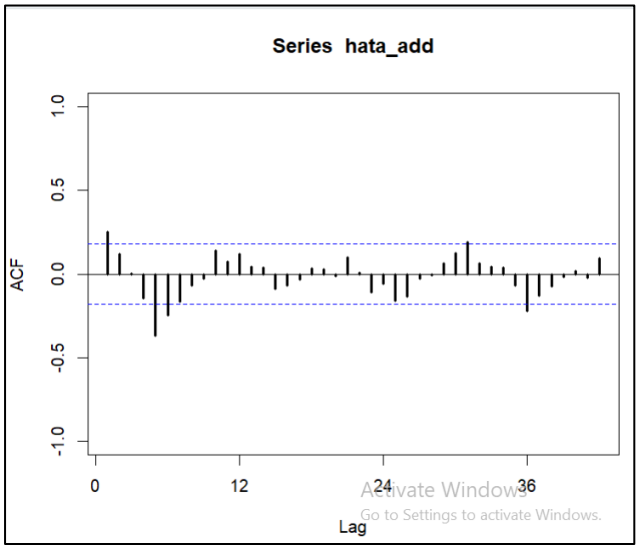
MASE = 0.296: The model performs quite well compared to the naïve forecasting method.

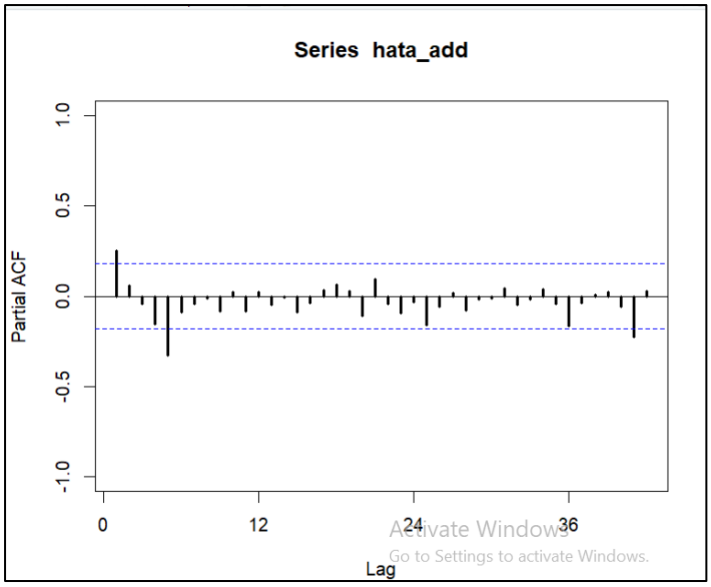
MAPE = 66.05%: The relative error is high; therefore, percentage-based error metrics have limited interpretability for this series.

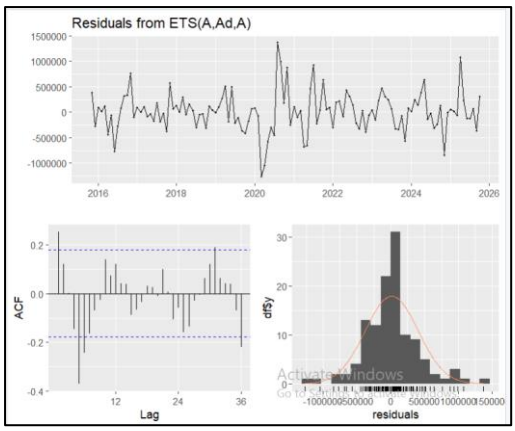
ACF1 = 0.252: There is positive autocorrelation in the residuals, meaning the residuals are not completely white noise.



```
> Box.test(hata_add, lag = 24, type = "Ljung")

        Box-Ljung test

data:  hata_add
X-squared = 53.383, df = 24, p-value = 0.0005149
```







Interpretation

Additive Holt–Winters: Original vs Fitted Series
When the graph is examined:

- The fitted series generally follows the original series closely.
- Both trend and seasonality are successfully captured.
- Some deviations are observed at peak and trough points.

Interpretation:
The model explains the overall structure well; however, forecast errors increase during periods with extreme values.

White Noise Assumption

- **ACF and PACF Plots:** Confidence bounds are exceeded at some lags.

Interpretation:
The residuals are not completely white noise; there is still some unexplained dependency structure in the series.

Ljung–Box Test (lag = 24)

- **χ^2 = 53.383**
- **p-value = 0.0005149**

Interpretation:
Since the p-value is less than 0.05, the null hypothesis (H_0: residuals are independent) is rejected. Thus, the residuals are not independent and the white noise assumption is not satisfied.

Overall Conclusion
The Holt–Winters additive (ETS(A,Ad,A)) model successfully captures the trend and seasonal structure of the time series and demonstrates strong forecasting performance compared to naïve methods. However, the fact that residuals are not fully white noise and the significant Ljung–Box test indicate that the model does not completely explain all aspects of the series. Despite this, the model is suitable for short- and medium-term forecasting.

Winters Multiplicative Method

```
> summary(Winters_mult)
ETS(M,A,M)

Call:
ets(y = visitors_ts, model = "MAM")

  Smoothing parameters:
    alpha = 0.3184
    beta  = 1e-04
    gamma = 2e-04

  Initial states:
    l = 2213797.0463
    b = 16533.4918
    s = 1.2422 1.5107 1.7386 1.7383 1.1725 1.0095
           0.7644 0.6011 0.5087 0.4846 0.5975 0.6319

  sigma:  0.2491

     AIC     AICc      BIC
3804.792 3810.792 3852.179

Training set error measures:
                    ME     RMSE      MAE      MPE     MAPE      MASE      ACF1
Training set -29215.36 568758.7 392043.4 -135.024 147.1045 0.4223173 0.6645578
```

Interpretation

Key Smoothing Parameters

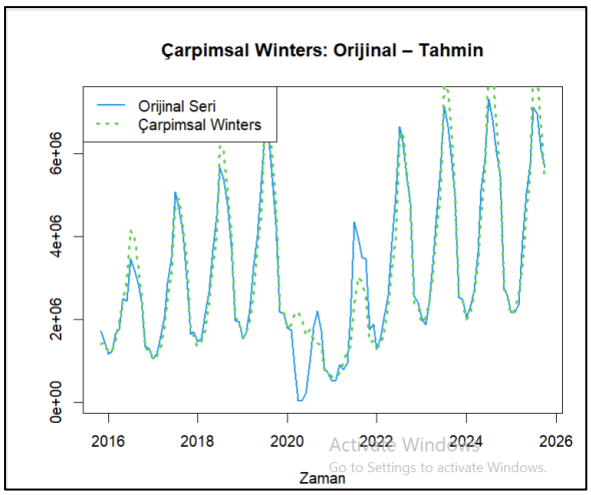
- **α = 0.3184:** The level component is updated at a moderate rate. The model is not overly sensitive to new observations.
- **β ≈ 0:** The trend component is nearly constant, indicating that the trend changes very little over time.
- **γ ≈ 0:** The seasonal component is assumed to be constant over time.

Interpretation:
The model largely assumes that both trend and seasonality are stable, incorporating new information only to a limited extent.

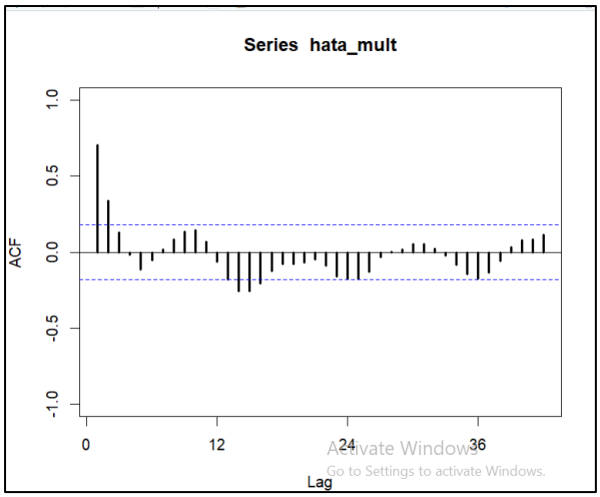
Seasonal Structure (Key Findings)
The fact that seasonal coefficients take values both below and above 1 indicates that in some periods the number of visitors is significantly above the average, while in others it is well below. This confirms the presence of strong seasonality.

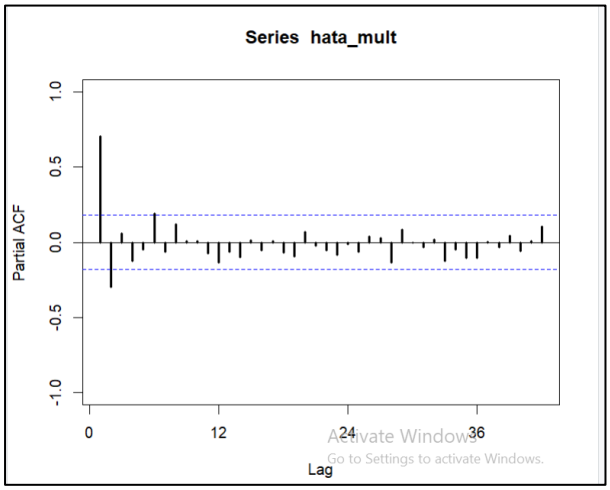
Model Performance (Key Metrics)

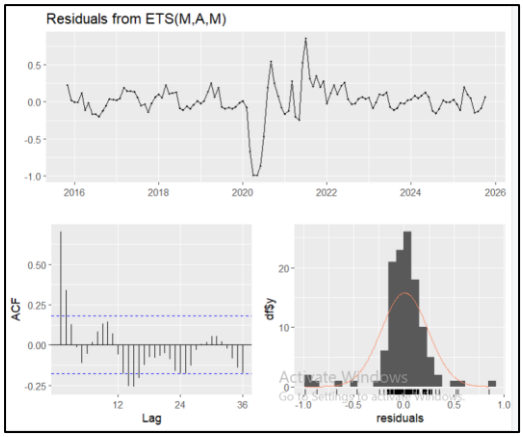
- **MASE = 0.422:** The model performs better than the naïve method, but worse compared to the additive model.
- **MAPE = 147.10%:** The relative error is extremely high, indicating that the multiplicative model does not adequately capture relative errors for this series.
- **ACF1 = 0.665:** There is strong positive autocorrelation among the residuals.




```
> Box.test(hata_mult, lag = 24, type = "Ljung")

        Box-Ljung test

data:  hata_mult
X-squared = 130.15, df = 24, p-value < 2.2e-16
```







Interpretation

Fit Between Original and Fitted Series When the graphs are examined, the model captures the overall seasonal structure; however, it shows significant deviations at peak and trough points. Compared to the additive model, the fitted series follows the original series more weakly.

White Noise Assumption Ljung–Box Test (lag = 24):

p-value < 2.2e−16

Interpretation: The null hypothesis (H_0: residuals are independent) is rejected. The residuals are not independent, and the white noise assumption is clearly violated.

Based on the joint evaluation of the ACF and PACF plots:

There is statistically significant autocorrelation in the residuals.

The residuals do not exhibit white noise behavior.

Overall Conclusion Although the Holt–Winters multiplicative model is able to capture the general seasonal structure of the time series, it shows substantial forecast deviations at peak and trough points. The fitted series follows the original series less closely compared to the additive model. The significant Ljung–Box test result and the presence of statistically significant autocorrelations in the ACF and PACF plots indicate that the residuals are not independent and do not satisfy the white noise assumption. These findings suggest that the multiplicative Winters model has limited forecasting performance for this dataset and does not fully capture the underlying dynamics of the series.

Comparing Models

```
> accuracy(SES)
                    ME     RMSE      MAE       MPE     MAPE      MASE     ACF1
Training set 30195.05 890851.3 684337.1 -26.6477 49.63044 0.7371822 0.466486
> accuracy(Holt)
                    ME     RMSE      MAE      MPE     MAPE      MASE        ACF1
Training set 2637.745 801793.8 652211.1 15.11145 50.70905 0.7025754 0.000957894
> accuracy(Winters_add)
                    ME     RMSE      MAE        MPE     MAPE      MASE      ACF1
Training set 13645.16 390090.7 274484.7 -53.25629 66.04825 0.2956806 0.2522586
> accuracy(Winters_mult)
                     ME     RMSE      MAE       MPE     MAPE      MASE      ACF1
Training set -29215.36 568758.7 392043.4 -135.024 147.1045 0.4223173 0.6645578
>
```

Box-Jenkins Modells

Definition of the Established Models

In this study, four different seasonal ARIMA models were estimated for the visitor series with monthly frequency (s = 12):

Model 1

SARIMA(0,1,1)(0,1,1)$_{12}$

Model 2

SARIMA(1,1,1)(0,1,1)$_{12}$

Model 3

SARIMA(0,1,1)(1,1,0)$_{12}$

Model 4

SARIMA(1,1,0)(1,1,0)$_{12}$

For all models:

- **d = 1:** First differencing is applied to remove the trend effect.
- **D = 1 (12-month):** Seasonal differencing is applied to remove seasonality.

Model 1: SARIMA(0,1,1)(0,1,1)$_{12}$

Coefficients

- **MA(1)**: significant ($p < 0.001$)
- **Seasonal MA(1)**: significant ($p < 0.001$)

This indicates that both short-term shocks and annual seasonal shocks are important in the series.

Goodness-of-Fit Measures

- **AIC = 3089.94**
- **Residual ACF(1) ≈ 0:** Residuals are largely random.

Interpretation:
The model has a simple structure, its coefficients are statistically significant, and it is consistent with the Box–Jenkins methodology. It serves as a fundamental and strong benchmark model.

Model 2: SARIMA(1,1,1)(0,1,1)$_{12}$

Coefficients

AR(1): significant ($p = 0.016$)

MA(1): significant ($p = 0.63$)

Seasonal MA(1): significant ($p < 0.001$)

Goodness-of-Fit Measures

- **AIC = 3088.74 (lowest)**
- **RMSE and MAPE are lower compared to the other models.**

Interpretation:
Although this model has the lowest AIC value, the insignificance of the MA(1) coefficient indicates that the model includes unnecessary parameters. This violates the principle of parsimony, which favors simpler models with only meaningful parameters.

Model 3: SARIMA(0,1,1)(1,1,0)$_{12}$

Coffecients

MA(1): significant

Seasonal AR(1): significant

Goodness-of-Fit Measures

- **AIC = 3098.28 (high)**
- **RMSE and MAPE have the highest values.**

Interpretation:
Although the seasonal structure is modeled using an AR term, the overall goodness-of-fit measures are weak. Compared to the other models, this model does not explain the series adequately.

Model 4: SARIMA(1,1,0)(1,1,0)$_{12}$

Coefficients

AR(1): significant ($p < 0.001$)

Seasonal AR(1): significant ($p < 0.001$)

Interpretation

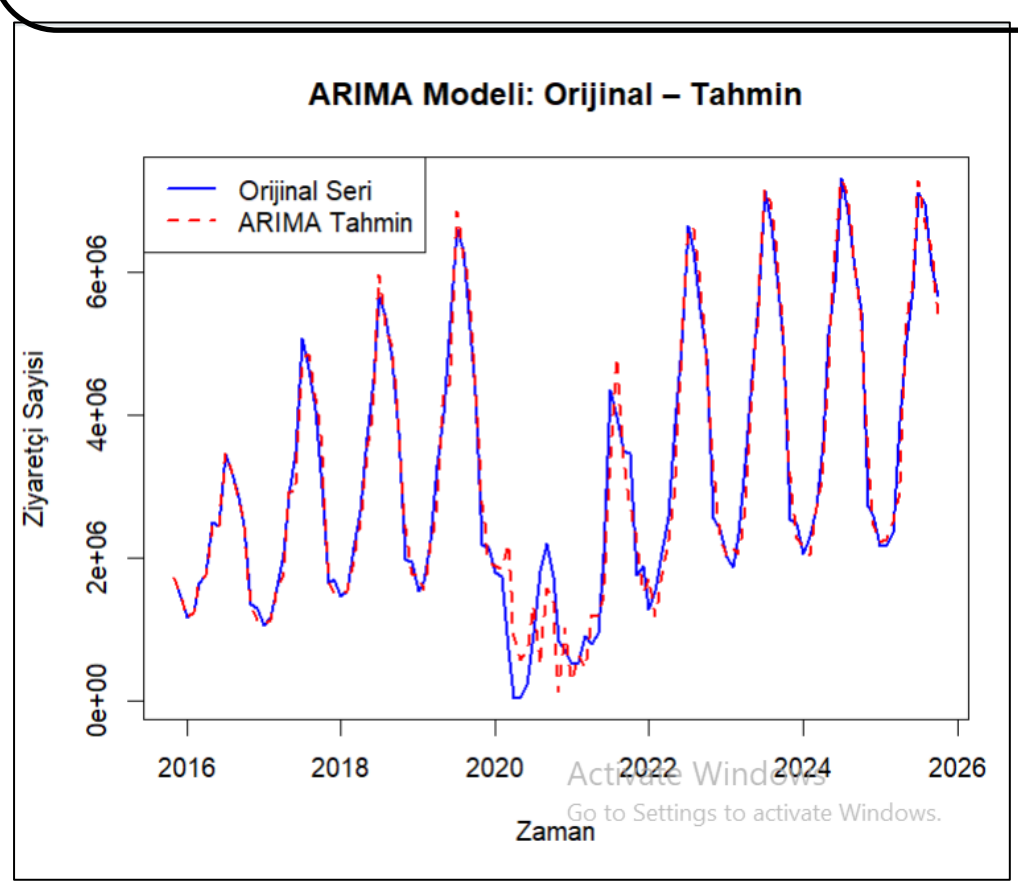
Goodness-of-Fit Measures

- **AIC = 3094.11**
- **Residuals exhibit low autocorrelation.**

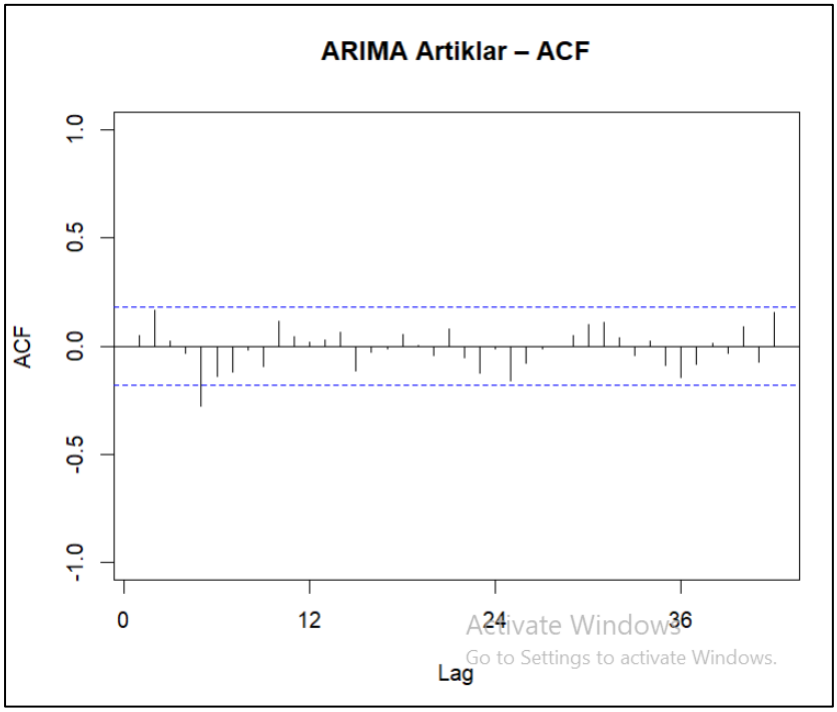
Interpretation:
The model includes only AR components, and all coefficients are statistically significant. However, its AIC value is higher compared to Model 1 and Model 2, indicating a relatively weaker fit.

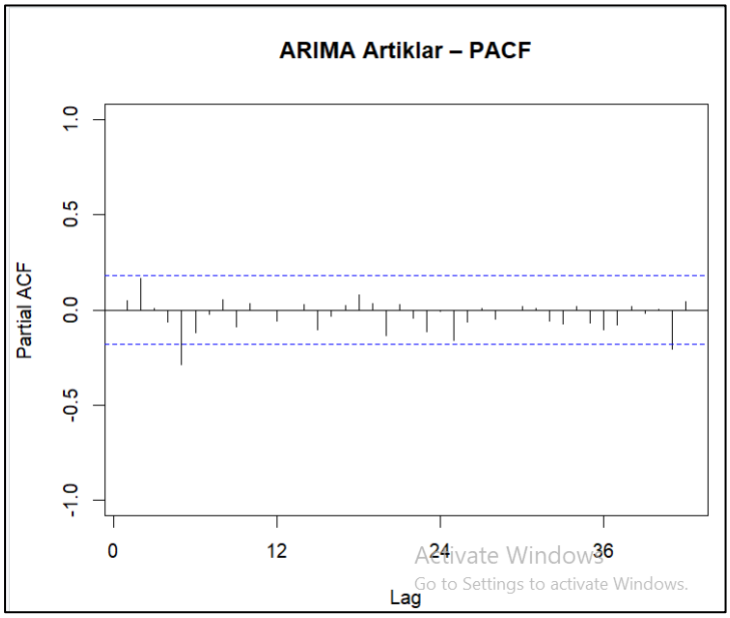
Final Conclusion

As a result of the Box–Jenkins analysis, four different SARIMA models were compared. Although Model 2 has the lowest AIC value, it was not considered parsimonious due to the statistical insignificance of the MA(1) coefficient. The **SARIMA(0,1,1)(0,1,1)12** model, in which all coefficients are statistically significant and which demonstrates a balanced performance in terms of goodness-of-fit measures, was selected as the final model, as it successfully captures both the trend and seasonal structure of the series.



```
> Box.test(res_arima,
+          lag = 24,
+          type = "Ljung-Box")

        Box-Ljung test

data:  res_arima
X-squared = 28.347, df = 24, p-value = 0.2457
```





Interpretation

Fit Between Original and Fitted Series
The fitted series (red) generally follows the original series (blue).
Seasonal fluctuations and trends are captured well by the model.
Although a small deviation is observed around 2020, there is overall strong agreement between the forecasts and actual values.

Interpretation:
This indicates that the model successfully explains the underlying structure of the time series.

Box–Ljung Test
The Box–Ljung test examines whether there is autocorrelation in the residuals.

- **χ^2 = 28.347**
- **df = 24**
- **p-value = 0.2457**

Interpretation:
Since p-value > 0.05, there is no significant autocorrelation in the residuals.
This means the residuals behave like white noise, indicating that the model has captured most of the information in the series. In other words, model assumptions are satisfied and residuals are randomly distributed.

ACF (Autocorrelation Function) Plot
The ACF plot shows correlations of residuals at different lags.
All autocorrelation values lie within the confidence bounds.

Interpretation:
There is no significant autocorrelation in the residuals, confirming that the model adequately captures the dependency structure. This is consistent with the Box–Ljung test result.

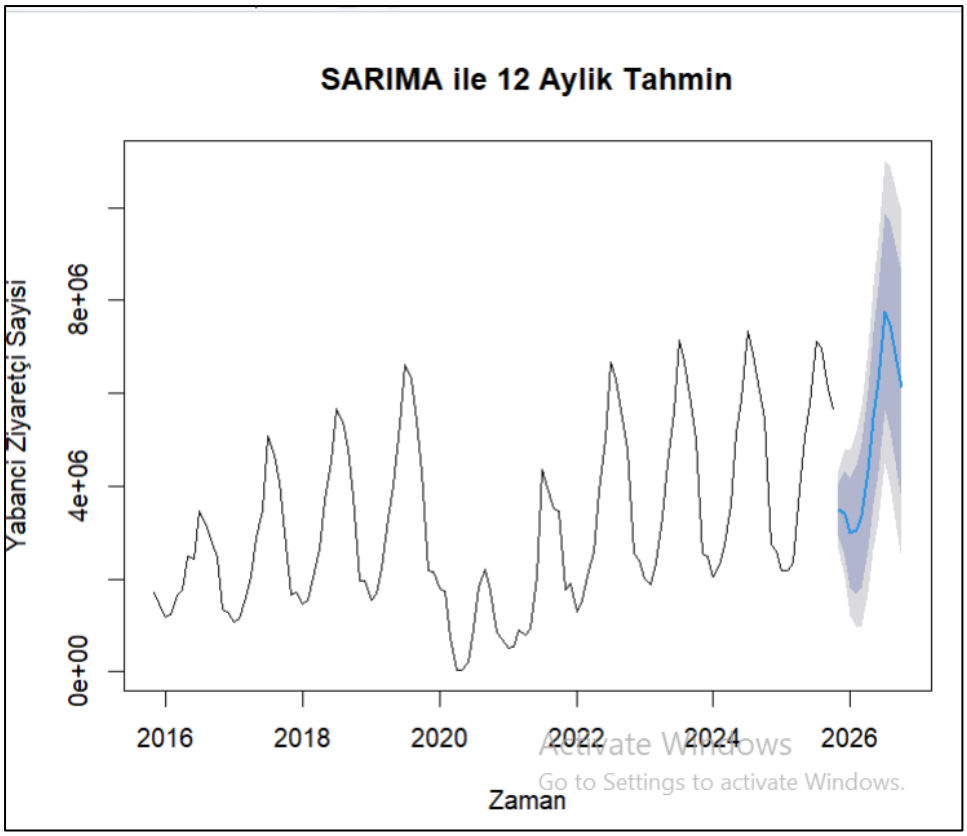
PACF (Partial Autocorrelation Function) Plot
The PACF plot shows the direct effect of each lag after removing intermediate effects.
All partial autocorrelations remain within the confidence bounds.

Interpretation:
There are no significant partial autocorrelations, supporting that the residuals behave like white noise.

Overall Evaluation

- **Model:** SARIMA(0,1,1)(0,1,1)[12]
- Forecasts are consistent with the original series, capturing both trend and seasonal components.
- Residuals behave as white noise, with no significant autocorrelation.
- Box–Ljung test, ACF, and PACF plots all support the validity of the model assumptions.

Final Conclusion:
The model is appropriate and can be considered a reliable forecasting tool for the time series.



12 Month estimation using SARIMA

```
+       ylab = "Yabanci Ziyaretçi Sayisi")
> forecast_arima$mean
         Jan     Feb     Mar     Apr     May     Jun     Jul     Aug     Sep     Oct     Nov     Dec
2025                                                                                    3491270 3406092
2026 2993160 3058236 3355547 4332560 5459090 6287591 7754309 7467890 6705063 6151800
>
```